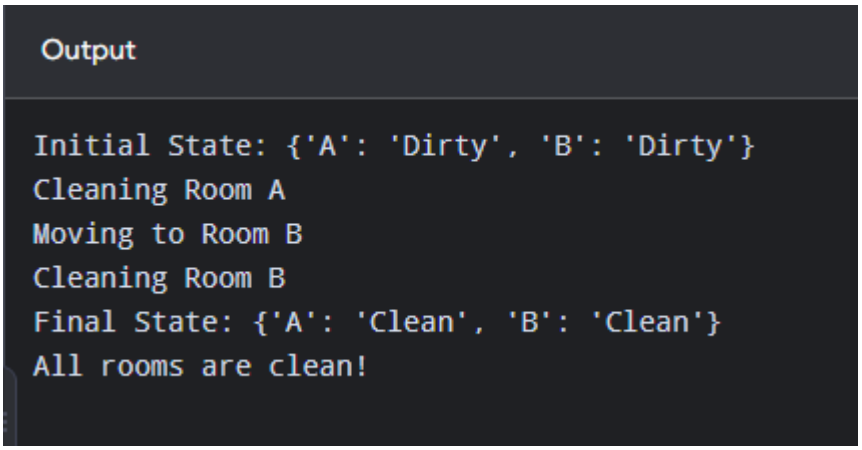
Lab experiment

Vacuum Cleaner Problem

Coding:

```
# Vacuum Cleaner Problem

# Initial state of rooms
rooms = {
    'A': 'Dirty',
    'B': 'Dirty'
}

# Initial position of vacuum cleaner
position = 'A'

# Vacuum cleaner function
def vacuum_cleaner():
    global position

    print("Initial State:", rooms)

    while 'Dirty' in rooms.values():

        if rooms[position] == 'Dirty':
            print("Cleaning Room", position)
            rooms[position] = 'Clean'

        elif position == 'A':
            position = 'B'
            print("Moving to Room B")

        else:
            position = 'A'
            print("Moving to Room A")

    print("Final State:", rooms)
    print("All rooms are clean!")

# Main program
vacuum_cleaner()
```

Output:


Output
Initial State: {'A': 'Dirty', 'B': 'Dirty'}
Cleaning Room A
Moving to Room B
Cleaning Room B
Final State: {'A': 'Clean', 'B': 'Clean'}
All rooms are clean!